Serial No. 368959

Original Instructions

OPERATING MANUAL

MAZATROL SmoothAi
MAZATROL SmoothX
MAZATROL SmoothG
MAZATROL SmoothEz5
MAZATROL SmoothEz
MAZATROL SmoothC

WORKPIECE THERMAL EXPANSION
COMPENSATION FUNCTION

Manual No.: H747SB0074E

Serial No.:

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Before using this machine and equipment, fully understand the contents of this manual to ensure proper operation. Should any questions arise, please ask the nearest Technical Center or Technology Center.

IMPORTANT NOTICE

1. **Be sure to observe the safety precautions described in this manual and the contents of the safety plates on the machine and equipment. Failure may cause serious personal injury or material damage. Please replace any missing safety plates as soon as possible.**
2. **No modifications are to be performed that will affect operation safety.**
3. **For the purpose of explaining the operation of the machine and equipment, some illustrations may not include safety features such as covers, doors, etc. Before operation, make sure all such items are in place.**
4. **If there is a difference between the contents of the manual and the actual state of the machine, please contact the nearest Technical Center or Technology Center.**
5. **Always keep this manual near the machinery for immediate use.**
6. **If a new manual is required, please order from the nearest Technical Center or Technology Center with the manual No. or the machine name, serial No. and manual name.**
7. **The export of machining programs may be restricted to prevent them from being used for military purposes. The machining programs you prepared are your property, so you should manage them yourself in accordance with the law.**
8. **Unauthorized copying or reproduction of this manual is prohibited.**

Issued by Manual Publication Section, YAMAZAKI MAZAK CORPORATION, Japan

03.2023

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1 SAFETY PRECAUTIONS

This manual is prepared only for the function or the device which is specified in the title of this manual. Therefore, you are required to carefully read and understand the operating manual of the machine, especially the section describing safety precautions, before using or operating the function or the device.

1-1 Rules

The meanings of our safety precautions to DANGER, WARNING, and CAUTION are as follows:

DANGER : Failure to follow these instructions could result in loss of life.

WARNING : Failure to observe these instructions could result in serious harm to a human life or body.

CAUTION : Failure to observe these instructions could result in minor injuries or serious machine damage.

1-2 Safety Precautions Relating to the Function/Device Described in This Manual

WARNING

- The barrier function does not work on the compensation operation performed by this function.
  The permissible amount of compensation is between +0.5 and –0.5 mm. The compensation operation within a range of +0.5 to –0.5 mm may be performed when the compensation enable/disable M-code (M786/M787) is executed, so the M-code must be executed when the workpiece is in a position where there is no danger of interference.
  Even for approach to the workpiece, a clearance of at least 0.5 mm should be left to prevent interference between the tool and the workpiece.

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2 OUTLINE

2-1 Intended Use

This function is designed to be used when the thermal expansion coefficient of the workpiece is higher than that of the machine (steel) and the room temperature of the factory is not 20°C.

When the room temperature during machining is not 20°C, the workpiece (aluminum) and the machine (steel) thermally expand in different ways, because they have different thermal expansion coefficients, with the result that a machining error may be involved, as shown in Fig. 2-1.

This function compensates for such a machining error and thus increases the machining accuracy. Therefore, it replaces the correction that has been made so far by experience or by adjusting the room temperature.

Note that when machining steel workpieces, there is no need to use this function, because steel workpieces thermally expand in the same way as the machine (steel).

Thermal expansion coefficient of aluminum materials: 23 (× 10^{-6}/°C)
Thermal expansion coefficient of the machine (steel): 12 (× 10^{-6}/°C)

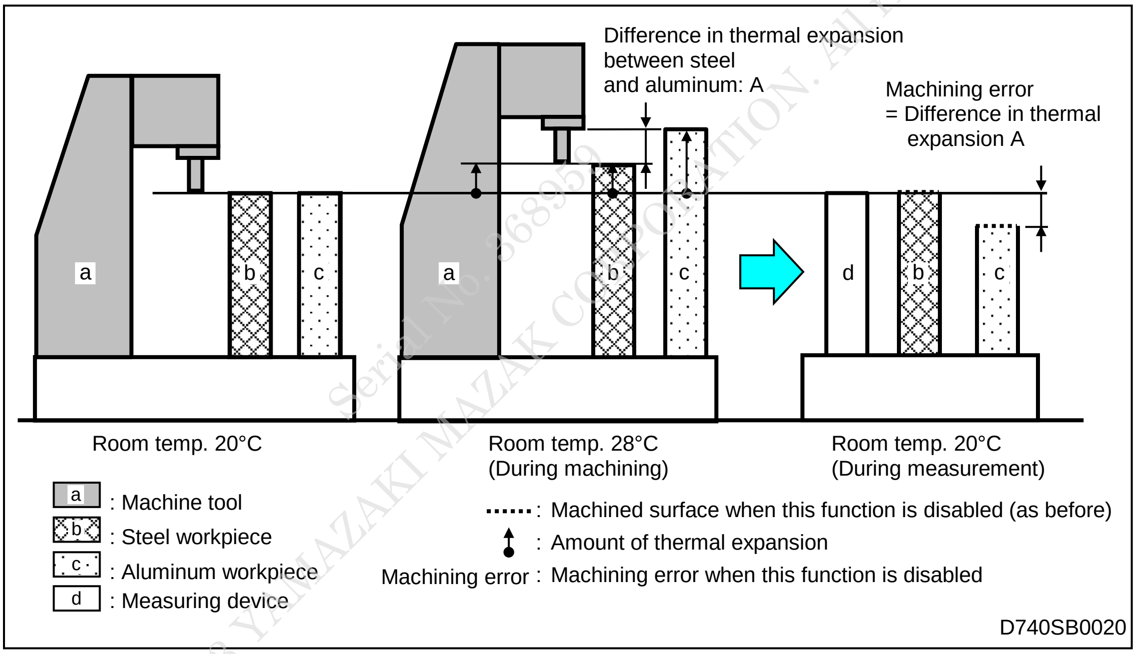

Fig. 2-1 Errors in the machining of steel and aluminum workpieces due to a difference in thermal expansion coefficient

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2-2 Compensation System

The amount of compensation is calculated from the machine tool temperature measured by the temperature sensor embedded in the machine and the correction factor, using the formula shown below. T_0 and **r** in this formula denote the arguments to be specified in the program.*1

*1 For the details of arguments, see 4-1, "Command formats and arguments".

Amount of compensation (μm)

$$= ((23 - 12)/1000) \times (\text{machine tool temperature} - T_0) \times \text{machining length (mm)} \times \frac{r}{100}$$

T_0: Temperature in the 3D measurement room (argument **f** or **d**)

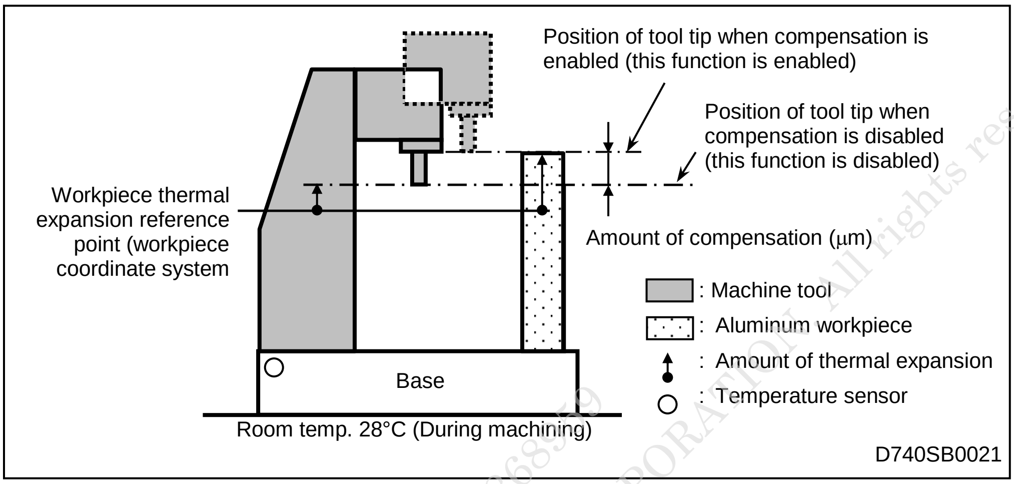

Fig. 2-2 Temperature sensor in the machine tool and amount of compensation

For example, for the machining of an aluminum workpiece, if the machine tool temperature is 30°C, the argument R is set to 100, d is set to 20°C, and the machining length is 600 mm, the amount of compensation will be +66 μm. The "machine tool temperature" here refers to the temperatures of the machine base.

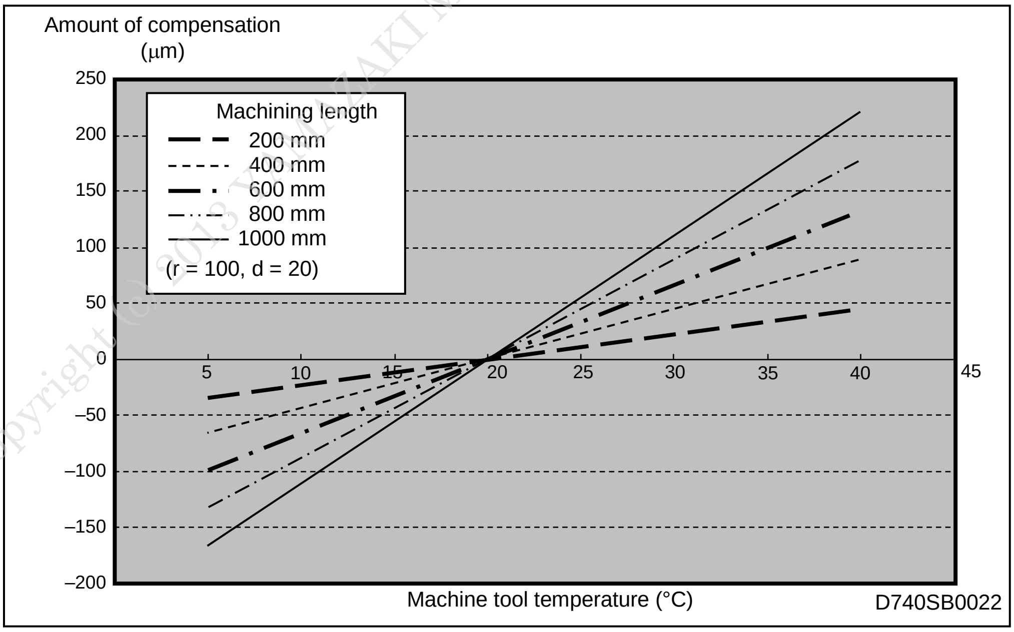

Fig. 2-3 Amount of compensation vs. machine tool temperature

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2-3 Example of Compensation

Here is an example of machining carried out with this compensation function enabled/disabled.

<An example of compensation>

Factory room temperature: 29°C
Machine tool temperature: 27.6°C
Measurement room temperature: 20°C
Product hole pitch: 500 mm
Dimensional tolerance: ±30 μm
Command value: 500 mm

Aluminum workpiece

- Measuring result from machining with compensation disabled:
  499.964 mm (–36 μm, outside the tolerance)
- Measuring result from machining with compensation enabled:
  500.006 mm (+6 μm, within the tolerance)

Steel workpiece

- Value measured with compensation disabled:
  500.003 mm (+3 μm)

Note : This function does not guarantee the machining accuracy, since the amounts of thermal expansion of the machine tool and the aluminum workpiece vary depending on how the workpiece is set up or how the machine is used.

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3 RESTRICTIONS

3-1 Restrictions

Restrictions imposed on the use of the thermal expansion compensation function are as follows.

1. The permissible amount of compensation is between +0.5 and –0.5 mm. This function cannot compensate for errors exceeding these limits.
2. Do not use this function when using an angle tool.
3. Compensation is made according to the amounts of movement on the X-, Y- and Z-axes. No compensation is made for movements of any rotational axis.
4. Do not use this function with any program that uses multiple-machining mode.
5. If two or more workpiece thermal expansion reference points are specified in an automatic operation program, and used for one and the same tool, do not use the identical-tool priority function for the tool.
6. This function uses G501 as the G-code to call a macroprogram, but do not use G501 in a manual program machining unit of MAZATROL program.
7. To enable compensation in the axial direction of the tool, register in advance the tool length (**LENGTH**) in the **TOOL DATA** display as shown in Fig. 3-1, Fig. 3-2 and Fig. 3-3. If the tool length is not registered, compensation will be carried out on the assumption that the tool tip is in the reference point of machine control, and so a machining failure may occur.

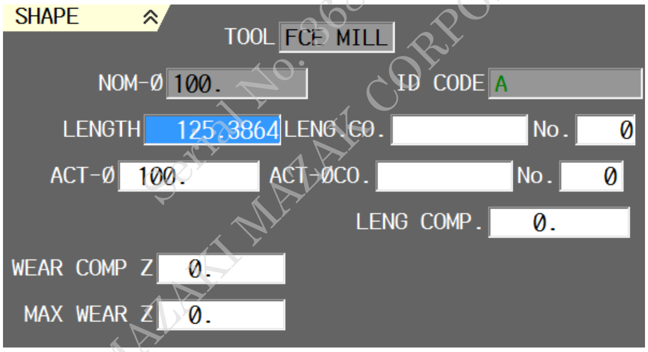

Fig. 3-1 Example of the **TOOL DATA** display (SmoothAi, X, G)

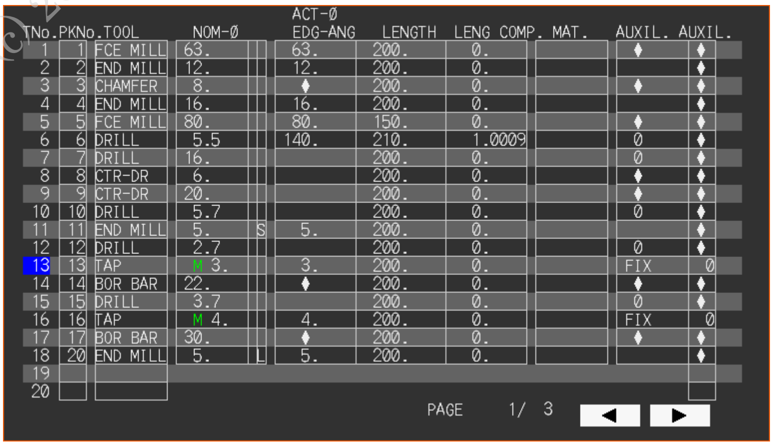

Fig. 3-2 Example of the **TOOL DATA** display (SmoothEz5, Ez)

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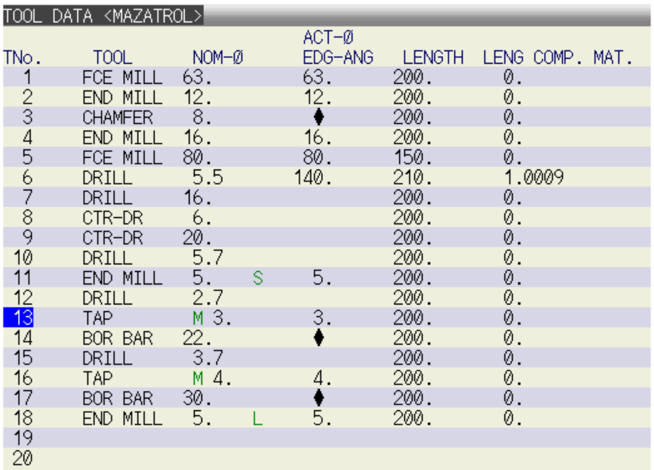

TOOL DATA <MAZATROL>

TNo.	TOOL	NOM-Ø	ACT-Ø EDG-ANG	LENGTH	LENG COMP.	MAT.
1	FCE MILL	63.	63.	200.	0.	
2	END MILL	12.	12.	200.	0.	
3	CHAMFER	8.	◆	200.	0.	
4	END MILL	16.	16.	200.	0.	
5	FCE MILL	80.	80.	150.	0.	
6	DRILL	5.5	140.	210.	1.0009	
7	DRILL	16.		200.	0.	
8	CTR-DR	6.		200.	0.	
9	CTR-DR	20.		200.	0.	
10	DRILL	5.7		200.	0.	
11	END MILL	5. S	5.	200.	0.	
12	DRILL	2.7		200.	0.	
13	TAP	M 3.	3.	200.	0.	
14	BOR BAR	22.	◆	200.	0.	
15	DRILL	3.7		200.	0.	
16	TAP	M 4.	4.	200.	0.	
17	BOR BAR	30.	◆	200.	0.	
18	END MILL	5. L	5.	200.	0.	
19						
20						

Fig. 3-3 Example of the **TOOL DATA** display (SmoothC)

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4 PROGRAM AND COMPENSATION OPERATION

4-1 Command Formats and Arguments

1. Command formats and arguments

This function can be executed by a special macro-call G-code with argument settings. Use M786 and M787 to enable and disable the compensation function, respectively.

```
G501XxYyZzRrDdFf;
M786; (Enables the compensation function)
 ⋮
Machining data
 ⋮
M787; (Disables the compensation function)
```

* A special manufacturer macroprogram, numbered 207400001, is called by a G501 command. Instead of G501, you may enter G65P207400001 here.

x: Specifies workpiece coordinate X of the workpiece thermal expansion reference point (system).

y: Specifies workpiece coordinate Y of the workpiece thermal expansion reference point (system).

z: Specifies workpiece coordinate Z of the workpiece thermal expansion reference point (system).

r: Specifies the correction factor in %. (Setting range: 0 to 500)
Specify 100 when machining an aluminum workpiece.

d: Specifies the room temperature of the 3D measurement room.
Setting range: 18.0 to 22.0 (Unit: °C)
Specify 20 in ordinary cases.
When specifying the argument Ff, omit Dd.

f: Specifies the room temperature of the 3D measurement room.
Setting range: 64.4 to 71.6 (Unit: °F)
Specify 68 in ordinary cases.
When specifying the argument Dd, omit Ff.

M786: Enables the compensation function.

M787: Disables the compensation function.

Note 1: Omit argument X, Y or Z when no compensation is desired for the particular axis.

Note 2: Argument address X, Y or Z can be entered without any value when the default one (0) is to be used to specify the workpiece thermal expansion reference point.

Note 3: To turn the compensation function back on during automatic operation after temporarily turning it off with the M787 command, enter the M786 command.

Note 4: Argument D or F can only be omitted when the default value (D20 or F68) is required.

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2. Workpiece thermal expansion reference point (in the workpiece coordinates system)

The “workpiece thermal expansion reference point (in the workpiece coordinates system)” to be specified with arguments refers to the datum point for thermal expansion of the workpiece. Therefore, no compensation is made for a positioning to that reference point. The amount of compensation that is determined by the distance from the “workpiece thermal expansion reference point” is added to the programmed distance of axis movement, as shown in Fig. 4-1.

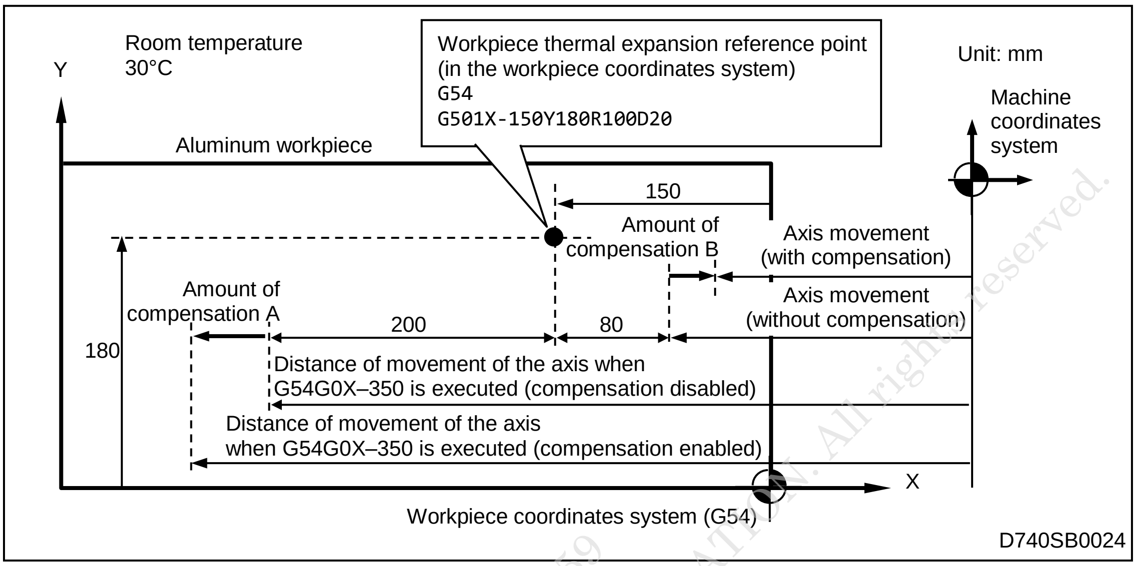

Fig. 4-1 Workpiece thermal expansion reference point and amount of compensation

Note 1: If a point away from the workpiece, for example, the machine zero point or the stroke end, is specified as the workpiece thermal expansion reference point, a machining failure may occur.
Therefore, the workpiece thermal expansion reference point should be set on the reference surface of machining or in the position of a locating pin.

Note 2: If the G501 command is executed several times, compensation is enabled only for the axes specified by the command executed last. To enable compensation for all the axes (X-, Y- and Z-axes), specify an argument for each of them.

Note 3: The workpiece thermal expansion reference point is to be specified with coordinates in the workpiece coordinates system that is active when G501 is executed.
If the workpiece coordinates system is changed to another while compensation is enabled and if you want to specify the workpiece thermal expansion reference point for the new workpiece coordinates system, execute the M787 command to temporarily disable the compensation function, and then execute the G501 command to specify the new workpiece expansion reference point again.

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3. Argument Rr

Various types of workpieces can be handled by changing the set value r of the argument R. The **r** for the thermal expansion coefficient of the workpiece to be machined can be determined from Table 4-1 or Fig. 4-2.

For example, when a workpiece with a thermal expansion coefficient of 40 × 10^{-6}/°C is machined by a machine with no scale feedback system, the set value **r** of the argument R is determined to be nearly equal to 250 from the graph in Fig. 4-2.

The **r** that can be determined from Table 4-1 and Fig. 4-2 is a theoretical value.

Table 4-1 Argument R set values for workpieces of different materials
(For machines with no scale feedback system)

Workpiece material	Thermal expansion coefficient (10^{-6}/°C)	Argument R set value (%)
Steel, cast iron	12	0 (No compensation)
Stainless steel	to 18	to 54
Brass	18	54
Aluminum	23	100
Plastic	from 23	from 100

For machines with scale feedback system

$$r = \frac{\alpha - 8 \times 10^{-6}}{11 \times 10^{-6}} \times 100\ (\%) \quad (\alpha\text{: thermal expansion coefficient of the workpiece})$$

For machines with no scale feedback system

$$r = \frac{\alpha - 12 \times 10^{-6}}{11 \times 10^{-6}} \times 100\ (\%) \quad (\alpha\text{: thermal expansion coefficient of the workpiece})$$

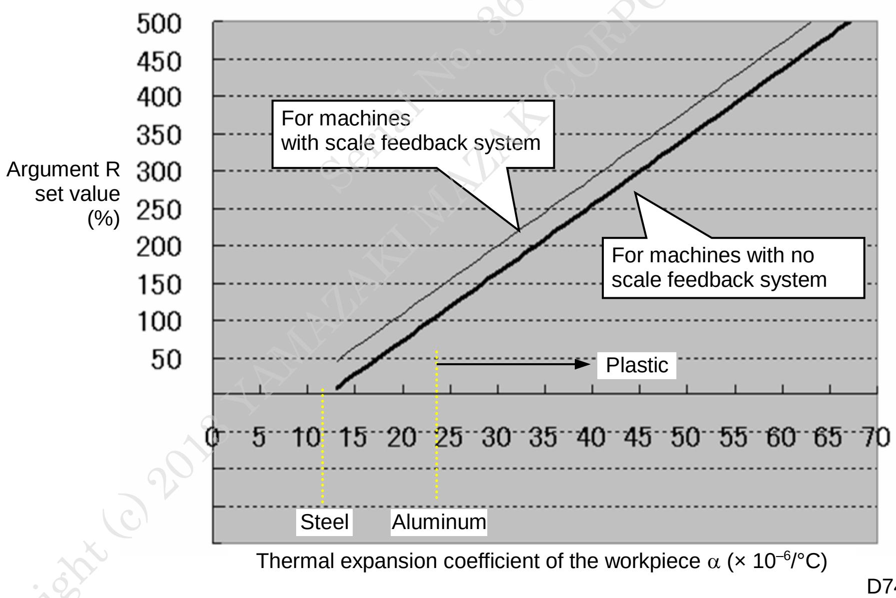

Fig. 4-2 Argument R set value and workpiece thermal expansion coefficient

Note : This function does not support workpieces with thermal expansion coefficients of less than 12 × 10^{-6}/°C.

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4-2 Compensation Operation

The position of the tool tip is corrected when the compensation enable/disable M-code is executed. If the machining is turned off or reset, or M30 is executed, the compensation function will be disabled and the setting of the workpiece thermal expansion reference point will be canceled. To enable the compensation function, specify the workpiece thermal expansion reference point and enter the M-code each time automatic operation is performed.

WARNING

- The barrier function does not work on the compensation operation performed by this function.
  The permissible amount of compensation is between +0.5 and –0.5 mm. The compensation operation within a range of +0.5 to –0.5 mm may be performed when the compensation enable/disable M-code (M786/M787) is executed, so the M-code must be executed when the workpiece is in a position where there is no danger of interference.
  Even for approach to the workpiece, a clearance of at least 0.5 mm should be left to prevent interference between the tool and the workpiece.

Note 1: Even after tool change for a touch sensor in the compensation mode, compensation will remain enabled. If necessary, disable the compensation function. During the measurement of the tool length, the detection of a break of a tool, an automatic tool change, etc., the compensation function is temporarily disabled, and it will be automatically enabled immediately after the completion of such an operation.

Note 2: The workpiece and machine coordinates indicated in the **POSITION** display do not include the amount of the compensation made by this function.

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4-3 Window for Compensation Enable/Disable Confirmation

The compensation enable/disable status can be confirmed in the status indicator window. The **W-THERM COMP** indicator in the status indicator window is lit when compensation is enabled, while it is off when compensation is disabled. (See Fig. 4-3 and Fig. 4-4.)

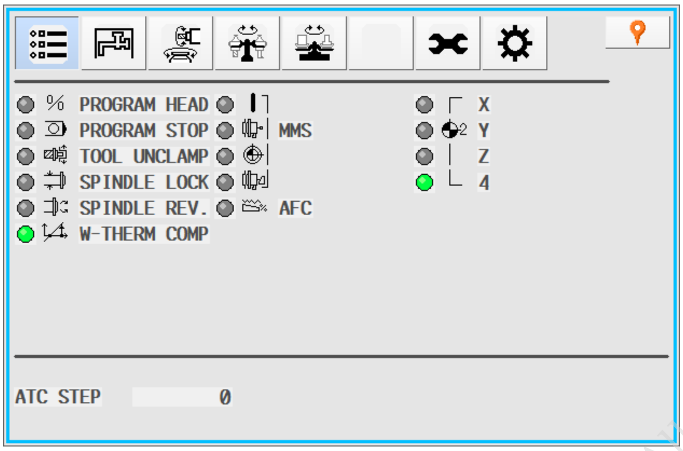

Fig. 4-3 Window for compensation enable/disable confirmation (example) (SmoothAi, X, G)

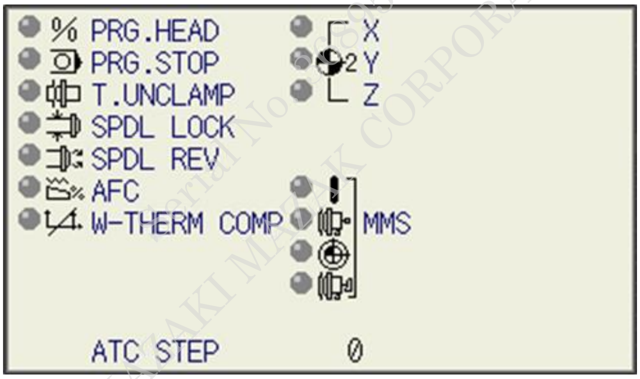

Fig. 4-4 Window for compensation enable/disable confirmation (example) (SmoothEz5, Ez, C)

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4-4 Examples of Programs and Explanation

Example of program 1: Hole machining in a flat aluminum plate (EIA/ISO)

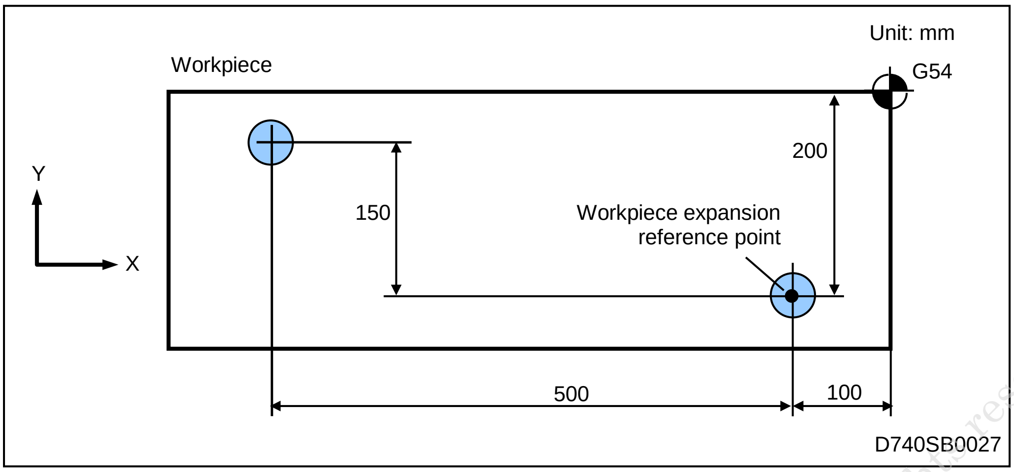

Fig. 4-5 Hole machining in a flat aluminum plate (EIA/ISO)

```
G90G94;
T06M6;
S300M3;
G54;
 ⋮ ;
N100;
G501 X-100.Y-200.R100.D20.;
```
.... Sets a workpiece thermal expansion reference point on G54.
Makes no compensation in the Z-axis direction.
3D measurement room temperature: 20°C

```
G0 X-100.Y-200.Z3.;
M786 ;
```
.... Enables compensation.
The amount of compensation is 0, because the tool is positioned at X–100 and Y–200.

```
G1 Z-30.F300.; (Boring)
G0 Z3.;
G0X-600.Y-50.;
```
.... Makes compensation according to a distance of movement of 500 mm in the X-axis direction and to a distance of movement of 150 mm in the Y-axis direction.

```
G1 Z-30.F300.; (Boring)
G0 Z3.;
M787;
```
.... Disables compensation.

```
M30;
```
.... Cancels the setting of the workpiece thermal expansion reference point.

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Example of program 2: Groove machining in a flat aluminum plate (EIA/ISO)

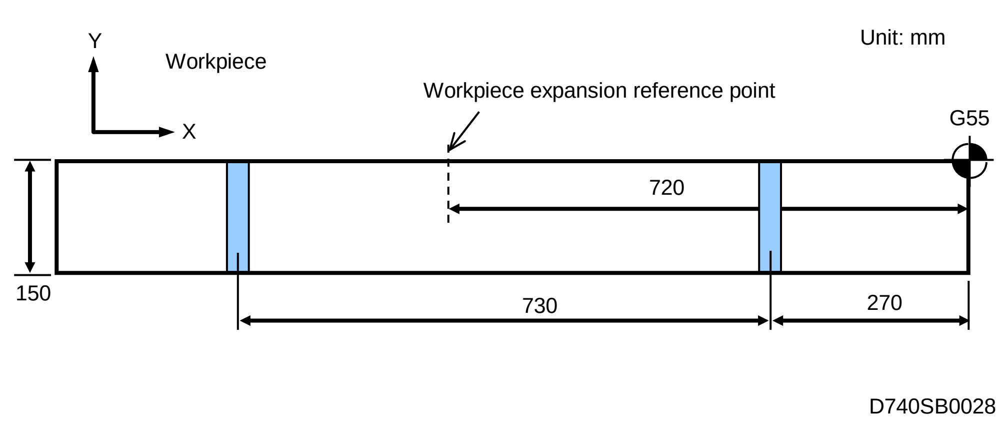

Fig. 4-6 Groove machining in a flat aluminum plate

```
G90G94;
T06M6;
S300M3;
G55.;
 ⋮ ;
N200;
G501 X-720.R100.D20.;        ............. Sets a workpiece thermal expansion reference
                                           point on G55.
                                           Makes compensation in the X-axis direction only.
                                           Makes no compensation in the Z-axis direction.
                                           3D measurement room temperature: 20°C
G0 X-270.Y-160.;
M786;                        ............. Enables compensation.
                                           Makes compensation according to the difference of
                                           450 mm between the workpiece thermal expansion
                                           coefficient reference point X (–720) and the
                                           command value X–270.
G0 Z-2.;
G1 Y10. F2000.; (Groove finish machining)
G0 Z50.;
G0 X-1000.Y-160.;            ............. Makes compensation according to the difference of
                                           280 mm between the workpiece thermal expansion
                                           coefficient reference point X (–720) and the
                                           command value X–1000.
G0 Z-2.;
G1 Y10.F2000.; (Groove finish machining)
G0 Z50.;
M787;                        ............. Disables compensation.
M30;                         ............. Cancels the setting of the workpiece thermal
                                           expansion reference point.
```

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Example of program 3: Hole machining in a flat aluminum plate (MAZATROL)

The manufacturer macroprogram 207400001 is called with a subprogram unit, and the workpiece thermal expansion coefficient reference point is specified with arguments, as shown in Fig. 4-8, Fig. 4-9 and Fig. 4-10.

In these examples, compensation is enabled in the X- and Y-axis directions, and the workpiece thermal expansion coefficient reference points is set to a point which is obtained by shifting the WPC origin (X = –131.0942, Y = –218.3159) by –100 and –200 in the X- and Y-axis directions, respectively.

Moreover, compensation is enabled for machining by centering drill, drill or reamer for which the compensation enable M-code M786 is entered.

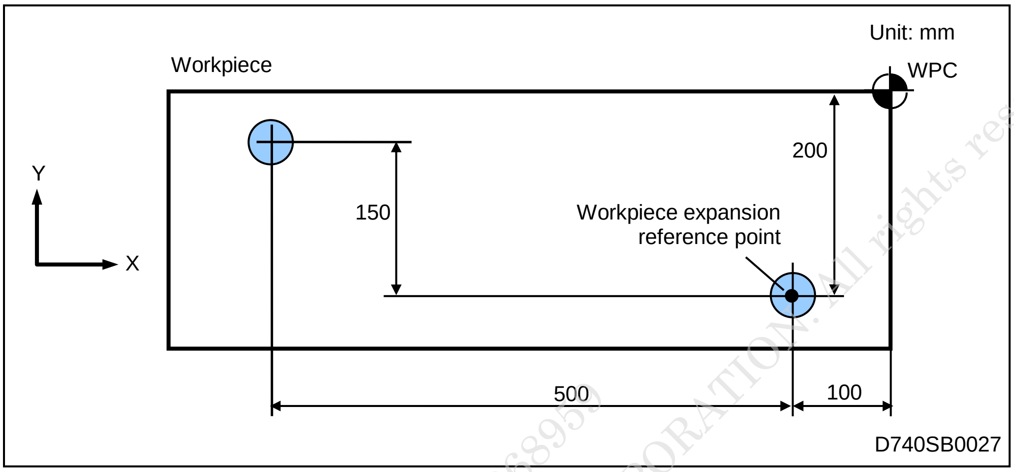

Fig. 4-7 Hole machining in a flat aluminum plate (MAZATROL)

UNo.	MAT.	INITIAL-Z	ATC MODE	MULTI MODE	MULTI FLAG	PITCH-X	PITCH-Y	DIS.
0	CST IRN	100.	1	OFF	♦	♦	♦	0

UNo.	UNIT	ADD. WPC	X	Y	th	Z	B
1	WPC- 0		-131.0942	-218.3159	0.	-219.6101	0.

UNo.	UNIT	WORK No.	$	REPEAT	No.
2	SUB PRO	207400001E	♦	1	
SNo.	ARGM 1	ARGM 2	ARGM 3	ARGM 4	ARGM 5	ARGM 6
1	X -100.	Y -200.	R 100.	D 20.		

UNo.	UNIT		DIA	DEPTH	CHMF	PRE-REAM	CHP
3	REAMING		10.	20.	0.	DRILL	0

SNo.	TOOL	NOM-Ø	No.	HOLE-Ø	HOLE-DEP	PRE-DIA	PRE-DEP	RGH	DEPTH	PK-DEP	C-SP	FR	M	M	M
1	CTR-DR	20. J		10.	♦	♦	♦	90°	CTR-D	♦	200.	0.1	786		
2	DRILL	9.8		9.8	21.	0.	100	DRIL	T 4.9	♦	150.	0.3	786		
3	REAMER	10.		10.	20.	♦	♦	♦	G01	♦	60.	0.05	786		

FIG	PTN	Z	X	Y	AN1	AN2	T1	T2	F	M	N	P	Q	R
1	PT	0.	-100.	-200.	♦	♦	♦	♦	♦	♦	♦	0	0	0
2	PT	0.	-600.	-50.	♦	♦	♦	♦	♦	♦	♦	0	0	0

UNo.	UNIT	CONTI.	NUMBER	ATC			RETURN		WORK No.	EXECTE
4	END									♦

Fig. 4-8 Example of program (SmoothAi, X, G) (MAZATROL)

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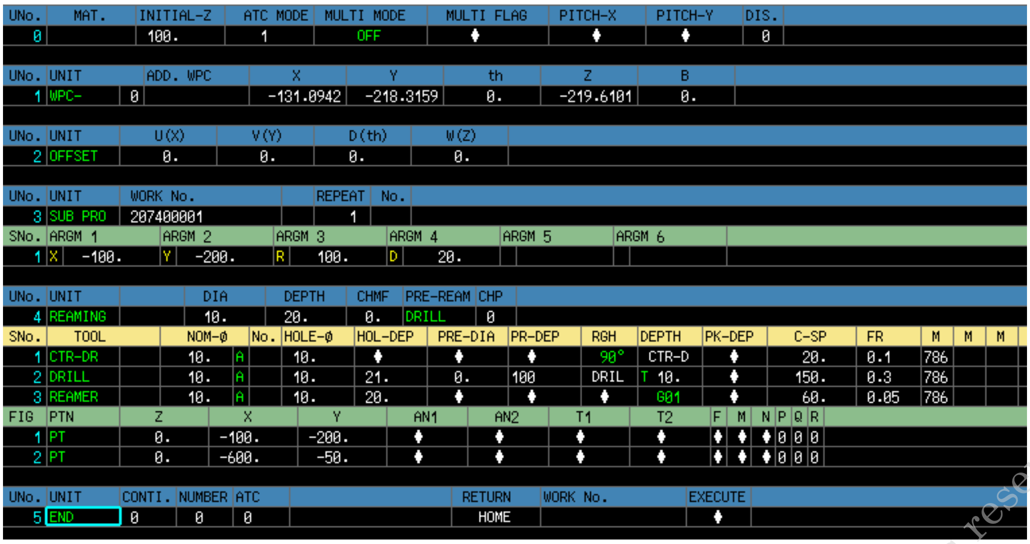

Fig. 4-9 Example of program (SmoothEz5, Ez) (MAZATROL)

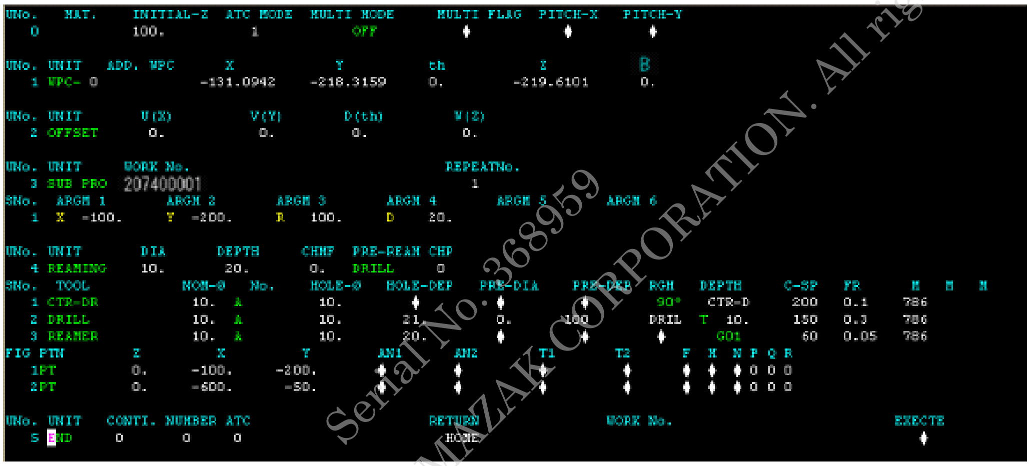

Fig. 4-10 Example of program (SmoothC) (MAZATROL)

Note : When centering drill, drill and reamer are all required for a machining unit, as shown in Fig. 4-8, Fig. 4-9 and Fig. 4-10, the compensation enable M-code M786 should be entered for each of them, otherwise a machining failure may occur.

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Example of program 4: Hole machining in a flat aluminum plate (MAZATROL)

The machining operations performed and the amount of compensation achieved by the programs shown in Fig. 4-11, Fig. 4-12 and Fig. 4-13 are the same as those by the programs shown in Fig. 4-8, Fig. 4-9 and Fig. 4-10.

The compensation function can also be enabled and disabled with M-code units, as shown in Fig. 4-11, Fig. 4-12 and Fig. 4-13, while in the examples shown in Fig. 4-8, Fig. 4-9 and Fig. 4-10, the M786 command is used to make a setting for each tool.

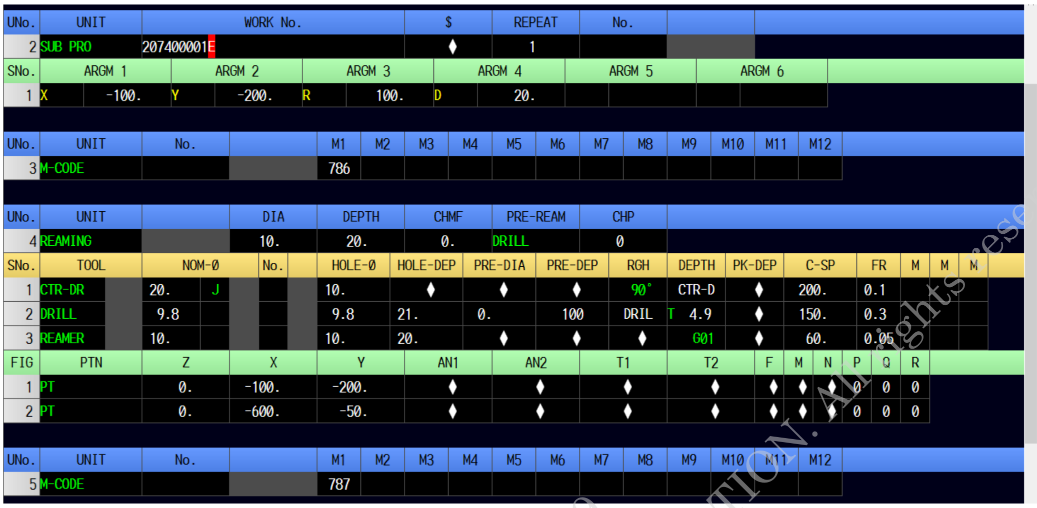

Fig. 4-11 Example of program (SmoothAi, X, G) (MAZATROL)

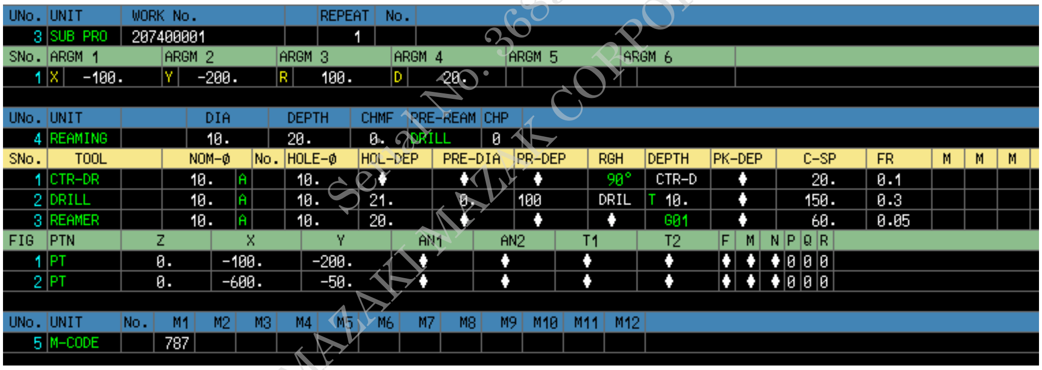

Fig. 4-12 Example of program (SmoothEz5, Ez) (MAZATROL)

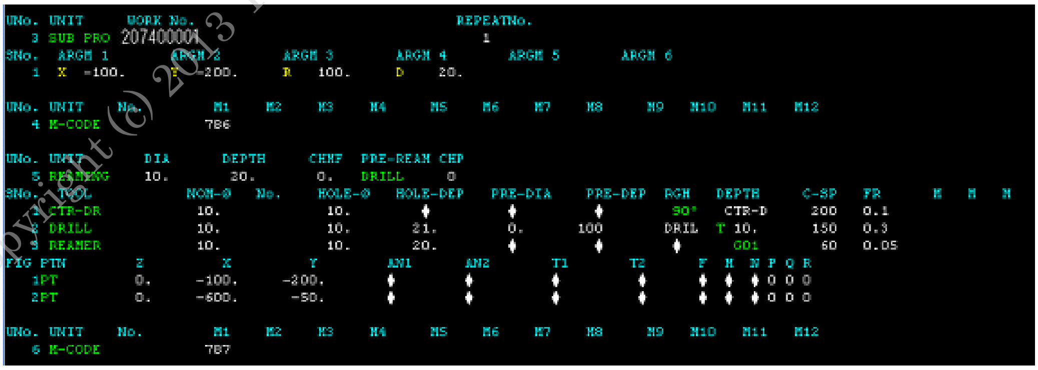

Fig. 4-13 Example of program (SmoothC) (MAZATROL)

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4-5 Caution When Restarting

1. If a workpiece offset G-code (G54 to G59) is used in an EIA/ISO or MAZATROL program

In the MDI mode, enter “G55” to specify the desired coordinates system, and “G501X-200.Y0. Z0. R100. D20.” to specify the workpiece thermal expansion coefficient reference point in that coordinates system, and then enter the compensation enable M-code M786, as shown in Fig. 4-14. After that, restart the operation without pressing the key, which enables compensation to start in the restart position.

If there is no need to enable compensation to start in the restart position, then it is unnecessary to enter M786 in MDI mode.

It is handy to prepare a subprogram that contains the items shown in Fig. 4-14, and to execute it in MDI mode. (Do not enter M30 in that subprogram.)

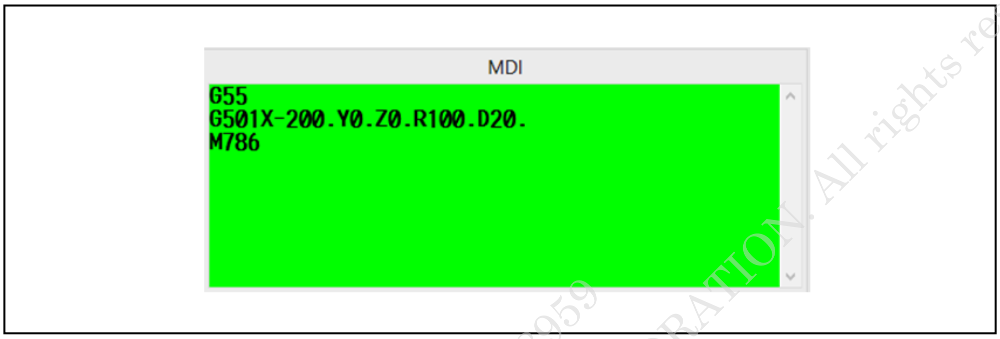

Fig. 4-14 Operation before restart (EIA/ISO program)

2. If a basic coordinates system (WPC) unit is used in the MAZATROL program

If a basic coordinates system (WPC) unit in which coordinates are specified is used, as shown in Fig. 4-15, Fig. 4-16 and Fig. 4-17, first transcribe the coordinate values to **WORK OFFSET** display.

Then, before the restart of operation, enter “G55” in MDI mode to specify the coordinates system concerned, enter “G501X-200.Y0. Z0. R100. D20.” to specify the workpiece thermal expansion coefficient reference point in that coordinates system, and then enter the compensation enable M-code M786. After that, restart the operation without pressing the key, which enables compensation to start in the restart position.

If there is no need to enable compensation to start in the restart position, then it is unnecessary to enter M786 in MDI mode.

It is handy to prepare a subprogram that contains the items shown in Fig. 4-15, Fig. 4-16 and Fig. 4-17, and to execute it in MDI mode. (Do not enter M30 in that subprogram.)

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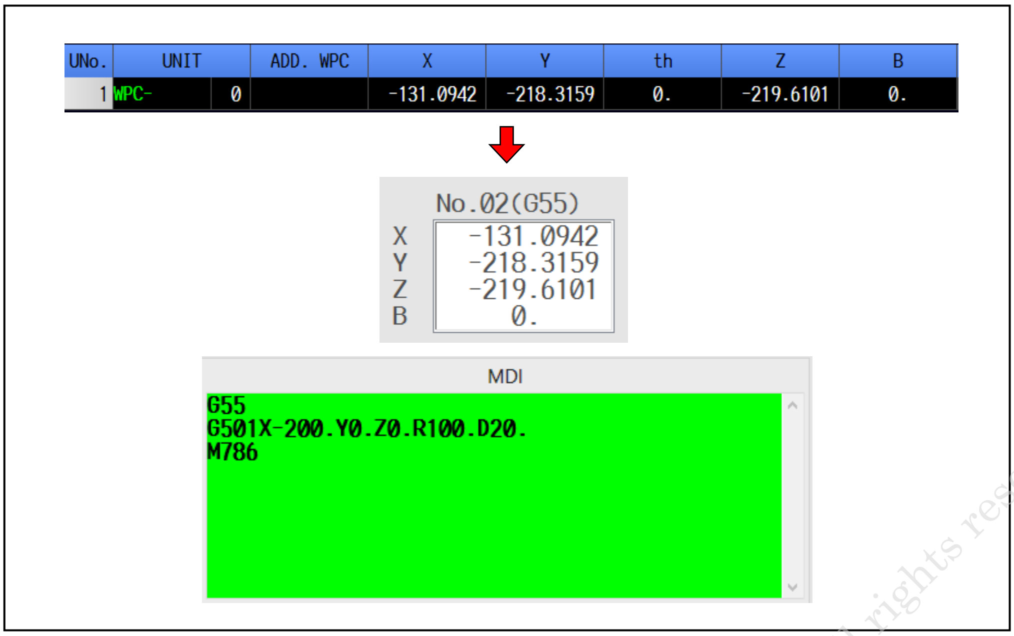

Fig. 4-15 Operation before restart (SmoothAi, X, G) (MAZATROL program)

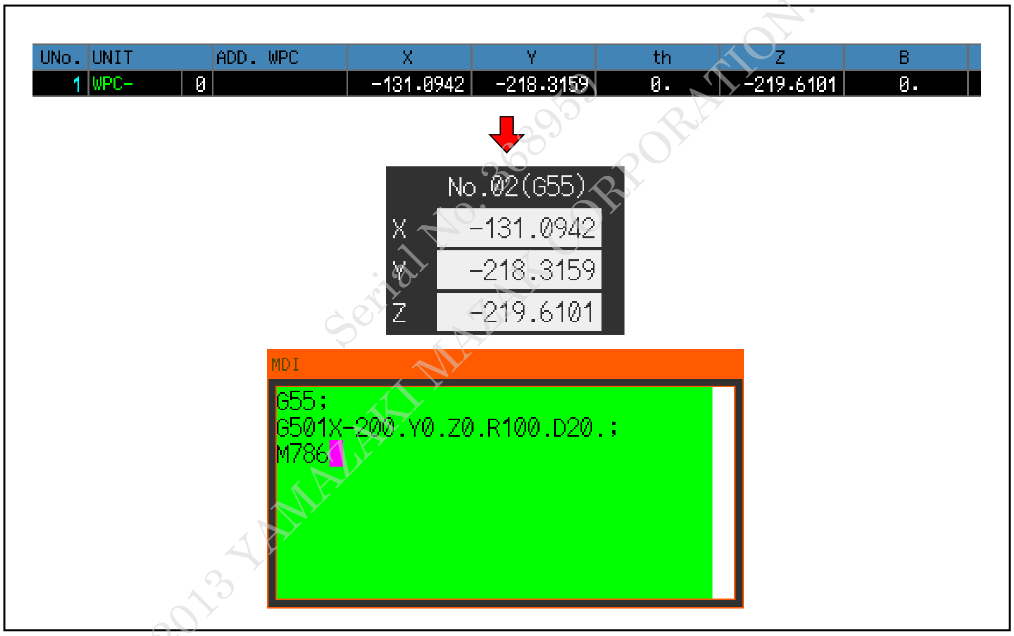

Fig. 4-16 Operation before restart (SmoothEz5, Ez) (MAZATROL program)

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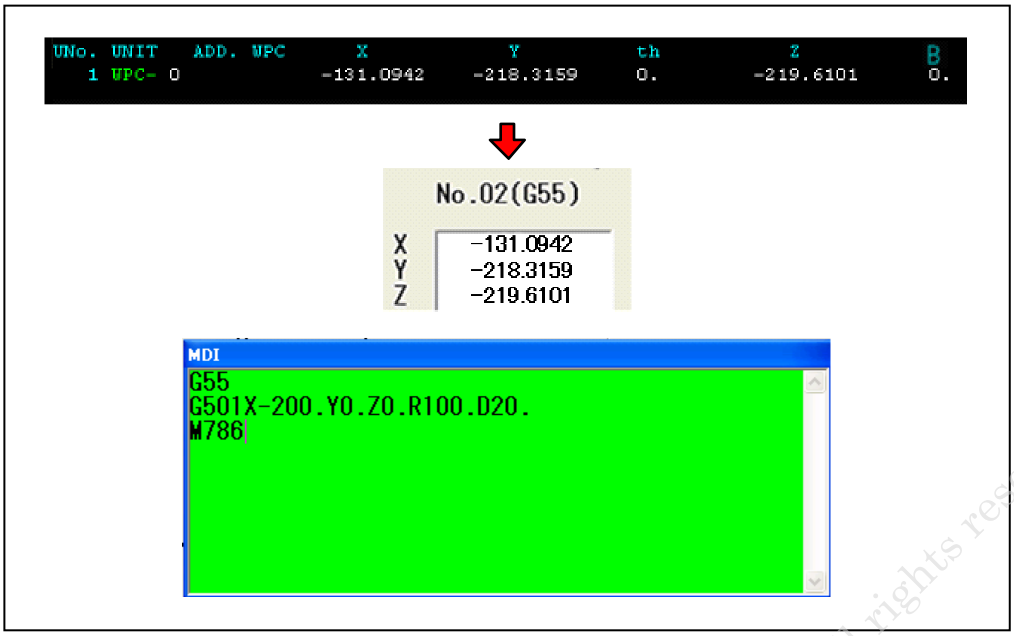

Fig. 4-17 Operation before restart (SmoothC) (MAZATROL program)

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5 ALARM, OPTION AND PARAMETER

5-1 Alarm

5-1-1 Structure of the alarm list

This alarm list is written in the following format:

No.	Message	Type of error	Stopped status	Clearing procedure	Display
[1]	[2] (, ,)	[3]	[4]	[5]	
Cause	[6]				
Action	[7]				

[1] Alarm number
[2] Alarm message
[3] Type of error

Code	Type	Description
A	Operation	A wrong key has been pressed. Or the machine has been operated incorrectly.
B	Registered data	The program or tool data includes an error(s).

[4] Stopped status

Code	Status
J	Single-block stop
L	Operation continued

[5] Clearing procedure

Code	Procedure
P	Press reset key
S	Press alarm clear button. (**Note**)

[6] Cause of alarm

[7] Action to be taken to eliminate the cause.

Note : For MAZATROL SmoothC, press clear key.

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5-1-2 Alarm list

No.	Message	Type of error	Stopped status	Clearing procedure	Display
186	W-THERMAL COMP. OVER LIMIT (, ,)	B	L	S	Blue
Cause	The amount of workpiece expansion compensation has gone beyond the limits of ±0.5 mm.				
Action	Specify a point on the workpiece as the workpiece thermal expansion reference point. Execute the compensate enable/disable command when the tool tip is in close proximity to the workpiece.				
187	SET W-THERMAL COMP. TOOL LENGTH (, ,)	B	L	S	Blue
Cause	While compensation in the axial direction of the tool was enabled, the tool was changed to another whose **LENGTH** was not specified in the **TOOL DATA** display.				
Action	Register **LENGTH** in the **TOOL DATA** display. See 3-1, “Restrictions”. Note that even if this alarm is issued, compensation is carried out normally in any direction other than the axial direction of the tool.				
188	SET W-THERMAL COMP. REF. POINT (, ,)	B	J	P	Red
Cause	Although the workpiece thermal expansion compensation reference point was not specified, the compensation enable command M786 was executed.				
Action	Specify the workpiece thermal expansion reference point.				

5-2 Option

No options are required.

5-3 Parameter

No parameters need to be set.

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6 SPECIFICATIONS

List of specifications of workpiece thermal expansion compensation function.

Item	Specification
1. Applicable NC	• MAZATROL SmoothAi • MAZATROL SmoothX • MAZATROL SmoothG • MAZATROL SmoothEz5 • MAZATROL SmoothEz • MAZATROL SmoothC
2. Applicable axis	X-, Y- and Z-axis (The value **LENGTH** in the **TOOL DATA** display tool is taken into account in Z-axis direction.)
3. Applicable program	• MAZATROL • EIA/ISO
4. Applicable workpiece material	Workpieces with thermal expansion coefficients of more than 12 × 10^{-6}/°C
5. Available range of compensation amount	–0.5 mm to +0.5 mm (In case of exceeding these limits, compensation amount is limited to ±0.5 mm.
6. Correction factor in %	0 to 500% (100% for machining an aluminum workpiece)
7. Number of reference points	1 for 1 axis (Only one reference point can be made valid for an axis.)

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